

Detection of Reckless Driving using Deep Learning

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Abstract—Reckless driving has always been a major concern for people. Every year it results in collisions and casualties. It is important to develop an automated system that detects reckless driving on highways. In order to overcome this issue, a new Deep Learning system is designed that detects rash driving and notifies the concerned authority to take necessary action. The proposed system has two main modules: (i) Extracting the trajectory and (ii) Detecting the dents in the vehicle body. Our proposed model demonstrates the promising results in detecting the rash driving. We validated our results using YouTube videos with promising accuracy.

Index Terms—Transfer_learning; Trajectory_Analysis; Dent Detection; Rash driving;

I. INTRODUCTION

A huge number of vehicles have been manufactured in the past decades to meet the customer's needs. The statistics show that since 2010 there will be an exponential increase in vehicle production. Such an extreme jump in a number of vehicles has caused serious issues for the current generation, it typically involves increased air/noise pollution, and road collisions in particular. World Health Organization [1] (Global status report on road safety 2018.) has shown that almost 1.25 million people (Figure 1) die each year from road traffic collisions. Most collisions are caused by poor driving habits (Figure 2). They include inappropriate operation, wrong judgment, etc. Drivers do not follow the rules of the lane in most situations, traffic regulations leading to traffic congestion and injuries. Good traffic situations and driver behavior [2] will prevent collisions and traffic congestion. Reckless driving [3] allows travel, transportation, and logistics to be a nuisance and it has a major impact not only on the rider and his / her family but also the passengers, drivers, and pedestrians, here the losses are of both life and money. Therefore, it is necessary to discover abnormal driving behaviors on the road to inform the concerned authority in order to prevent such activities. As per state-of-art, there is a need to develop a satisfactory, effective, and cheap method to detect the rash driving in real-time.

In this work, we propose a system using deep learning techniques which consists of two modules for detecting rash

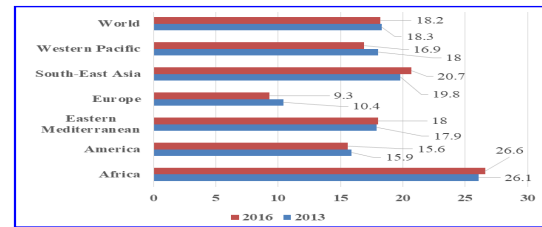


Fig. 1. Road traffic death rates for WHO regions per 100K population: 2013, 2016. Global road traffic death rates are 18.2 per 100K citizens, with major differences across regions of the world, varying from 9.3 to 26.6 per 100K citizens.

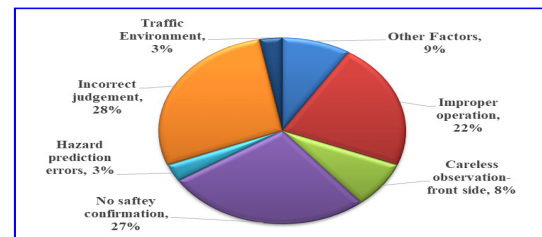


Fig. 2. Traffic Incidents based on Driving behaviors [4]. Violations of traffic law resulting in fatal traffic collisions as published by the NPA (Japan).

driving automatically. Module 1: Here we hypothesize that dents on the vehicle can predict the behavior of the driver. In this module, we trained a deep neural network using images that contains the vehicle has dents and scratches, which is appeared after collision. For the given image this module predicts the probability of dent which directly infers the rash driving behavior of the driver. Module 2: In this module, we extract the trajectory of a vehicle from the sequence of frames in the given video. Then we use this trajectory to predict the rash driving by using the model trained. By considering the probabilities from both modules we are able to detect the rash driving with decent accuracy. The remainder of the paper will be structured as follows. The summary of the related work is described in Section II, followed by the description of the

proposed methodology in Section III. Section IV explains the experimental results and discussions. Section V concludes the work with direction for future work.

II. RELATED WORKS

In recent years, smartphone-based event detection in the construction of a smart transport system has been a quiet area of research. For rash driving detection, dedicated sensors such as on-board-device (OBD) apps inside cars, smartphone-based mobile applications or roadside sensors have been implemented in most previous methods. Driving event detection [5] using data collected from GPS receiver and smartphone accelerometer, conveniently without the assistance of other external sensors or hardware. Sensory data collected by smartphones are typically noisy, resulting in an unreliable measurement of driving behaviors. A robust smartphone autocalibration-based driving behavior recognition scheme - DrivingSense [6] determines the sensor noise distribution while driving a vehicle. DrivingSense leverages the modified sensor parameters to classify three forms of dangerous behaviors: speed, frequent change of direction and excessive control of speed. Artificial neural network (ANN) based techniques [7] have been widely used to predict vehicle speed that helps improve engine performance in the future. A deep learning architecture [8], [9], [2] can learn the essential features from the raw data automatically and detects the surface anomalies. A deep auto-encoder learning model was proposed [10] to analyze bearing failures. Cluster sequence mining (CSM) [11] which selects the cluster pairs with occurrence correlation from event sequence data. This CSM is used to evaluate damage interactions in a fuel cell. A deep learning-based solution for classifying car damage [12], the model involves pre-training focused on Convolution Autoencoder followed by supervised fine-tuning, transfer learning, integrated learning process, and notable accuracy on dent recognition has been achieved.

An effective method to identify low-illumination and small-scale traffic signs ([13]), where Dark Area Sensitive Tone Mapping (DASTM) technique is used to enhance the illumination of dark image regions with little influence on bright regions. An artificial neural network-machine vision-based model [14] that recognizes and interprets the road and traffic sign of Malaysia in real-time. A genetic (machine vision) algorithm [15] that detects traffic violations and helps to clearly sweep and block the pedestrian lane. It is quite expensive to install loop detectors, roadside speed indicators, constant vigilance through CCTVs to throughout the city. There was a great need for a cost-effective and real-time road safety monitoring solution [16] since the smart transportation system is one of the most important components for a smart city.

III. METHODOLOGY

The proposed reckless driving detection system has two main modules. The first module is to determine the damage (detection of dents) on the vehicle. The second module

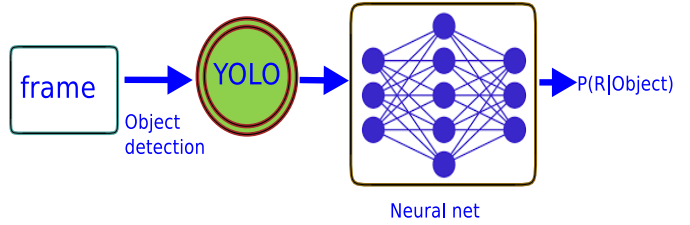


Fig. 3. The graphical outline of module 1 (Vehicle damage detection). Rash driving detection by considering dent /damage on the object detected by YOLOv3 ([18]) in the given video frame.

addresses the undesired vehicle trajectory. For the given video footage at first, the objects are identified using a multi-object detection algorithm (YOLOv3) [17], later the damage/dents on the previously recognized objects (vehicles) are identified using the trained model as explained in section III-A. Similarly for the trajectory of the vehicle movement is identified using the model as explained in section III-B. Then in section III-C, we have estimated the probability of rash driving using the probabilities given by module 1 and module 2.

A. Vehicle damage detection

We trained a deep neural network in this module using images that include the vehicle with dents and scratches that occur after a crash or a collision. For the given image this module (Figure 3) predicts the probability $P(R|Object)$ of rash driving. Deep learning has shown promising results in image classification applications. Transfer learning is a well-known technique that has worked effectively in the case of small labeled data. A transferable feature representation learned by CNN minimizes the effect of overfitting. The trained weights of the VGG16 network can be used effectively to project the image into some feature space. Further, this projected feature can be used to train the fully connected neural network as shown in Figure 5. In the first part of the architecture, we are projecting the image of the vehicle to a feature space by using a pre-trained sub-network of VGG-16. The main objective to project the image into featured space using a deeper network ensures the availability of more information in the projected space which makes training more accurate and ensures fewer trainable parameters. Hence, it effectively reduces the complexity of the trainable network and takes less training time.

Our CNN architecture consists of 13 layers: Conv1-Conv2-MaxPool1- Conv3-Conv4-MaxPool2- Conv5-Conv6-Conv7-MaxPool3- Conv8-Conv9-Conv10-MaxPool4 where Conv and MaxPool denote Convolutional layer and Max pooling layer respectively. The convolutional and the pooling layer form a pre-trained (non-trainable) sub-network of VGG16. The output generated using VGG16 has 25088×1 dimensions. $224 \times 224 \times 3$ Image has converted into 25088×1 dimensions. The features of the image are converted into a feature vector. The output of this sub-network is fed as input to the second half of the model which consists of two hidden layers followed by the sigmoid output layer (Figure 5). The first hidden layer

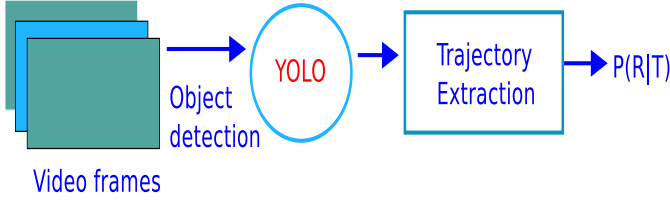


Fig. 4. The graphical outline of module 2 (Trajectory extraction and analysis). Extracting trajectory from the sequence of frames in a given video.

has 80 nodes and ReLU as the activation function. It has $25088 \times 80 + 80$ trainable parameters. And the second hidden layer has 40 nodes which also has ReLU as activation function and having $80 \times 40 + 40$ trainable parameters. The Sigmoid output layer has $40 + 1$ trainable parameters. The two fully connected layers and the sigmoid output layer are trained using the original as well as an augmented dataset. This type of network can be trained using a small dataset in less time. And it also prevents the overfitting problem. From this layer, we are calculating the probability of threat given that the vehicle is damaged which will be further used to calculate the final probability.

B. Trajectory analysis

In this module, we extract a vehicle's trajectory from the frame sequence, then we use this trajectory to predict rash driving probability $P(R|T)$ in the given video as shown in Figure 4. Due to their rapid response to high accuracy and strong adaptability, computer vision techniques have become important in real-time applications. The main objective of this phase is detecting and locating the vehicle in successive frames of a video and trace its path. The tracking algorithms are faster than the detection algorithm because, an object that was detected in the previous frame contains a lot of information such as its location, direction, and speed of its motion. So we can use all this information to understand the trajectory. A vehicle traveling with multiple lane changes in a short period conducts an S-shaped path. In this type of path, the direction of the vehicles changes so frequently and so the slope. These types of motions are unusual for the normal drivers which cause confusion in the mind of other drivers for the next movement. If a change in the direction and change in the degree of slopes crosses the threshold multiple times then it will be inferred as rash driving. Figure 8 describes the formed trajectory of a vehicle movement (ref. Figure 7) by considering frameset as shown in Figure 6. This form of reckless driving is therefore identified as an exception as it poses a danger in video surveillance systems to other traffic subjects.

For classifying the trajectory as bad or good we have mapped the estimated theta (θ) to the probability between 0 and 1 by using the following function (a function for mapping estimated theta to probability) (Please ref Figure 9).

$$f = P(R|\theta) = \begin{cases} \sin \theta & \text{if } 0 \leq \theta \leq 90^\circ \\ 1 & \theta > 90^\circ \end{cases}$$

Theta (θ) estimation : For any three frames, we have taken a coordinate position given by YOLO ([18]) for the given vehicle. We made a coalition of all three frames to form two vectors $\vec{a}$ and $\vec{b}$ respectively by subtracting the coordinates in the previous frame from the coordinates in the later frame as illustrated in Figure10. To estimate theta (θ) we use the following formula.

$$\theta = \cos^{-1} \left(\frac{\vec{a} \cdot \vec{b}}{||\vec{a}|| \cdot ||\vec{b}||} \right)$$

Here we are considering only horizontal axis susceptible movements to find the probability of a threat. From the video, it will capture frames per second and will do image processing for those captured frames. Using YOLOv3 ([18]) the vehicles will be detected and localized, the points of the diagonal element will be used to find the center for each vehicle. YOLOv3 uses Darknet-53 as a feature extractor and it does not use any form of pooling. It uses a convolutional layer with stride 2 to downsample the feature maps. So the loss of information is less in this case. The squared bounding box framed outside the object gives the diagonal co-ordinates. The center of the bounding box will be found using these two points. Plot the center for each vehicle on the graph. By using this we can trace the path of the vehicle.

C. Considering the two modules for final prediction

To generate final predictions (detection rash driving) we consider two major parameters which are objects and trajectory of the vehicle movement. Here, $P(R|Object)$ the probability for rash driving given dent and $P(R|T)$ the probability of rash driving given trajectory is estimated by module 1 and module 2 respectively. Now, using the Bayesian approach we will calculate the probability of rash driving given both extracted trajectory and the damage on the vehicle. We combine these two factors together under the assumption of independence.

$$P(R | D, T) = \frac{P(D, T | R)P(R)}{P(D, T)} \quad (1)$$

$$= \frac{P(D | R)P(T | R)P(R)}{P(D)P(T)} \quad (2)$$

$$= \frac{\frac{P(R | D)P(D)}{P(R)} \frac{P(R | T)P(T)}{P(R)} P(R)}{P(D)P(T)} \quad (3)$$

After simplification we get

$$P(R | D, T) = \frac{P(R | D)P(R | T)}{P(R)} \quad (4)$$

where $P(D)$ is the probability of dent and $P(T)$ is the probability of the trajectory. Figure 11 clearly describes the operations involved in the validation process of the proposed reckless driving detection model.

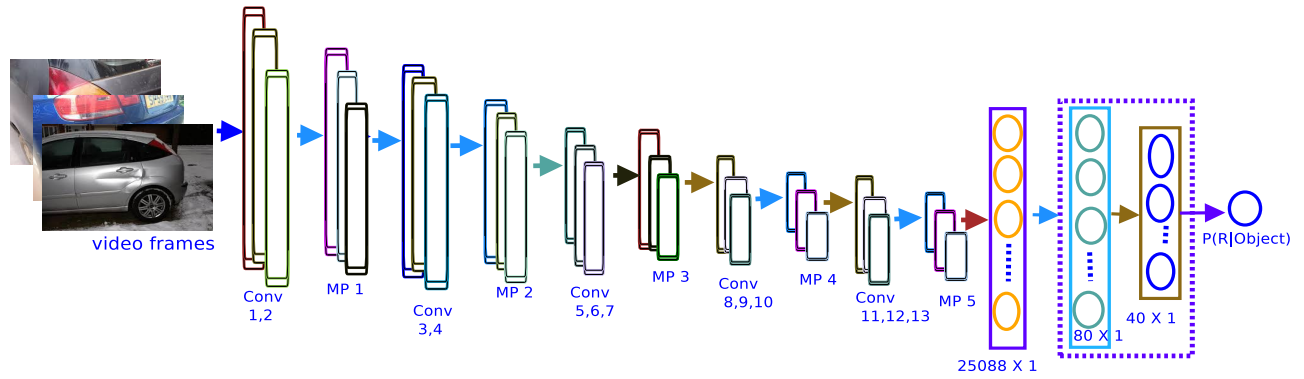


Fig. 5. Damage Detection Architecture (Trainable Neural Network). The initial (hidden) layer has 80 nodes, and ReLU is used as an activation function, it has trainable parameters of $25088 \times 80 + 80$. A total of 40 nodes are available in the second hidden layer and $80 \times 40 + 40$ trainable parameters are available for the ReLU activation function. There are $40 + 1$ trainable parameters in the output layer (Sigmoid).

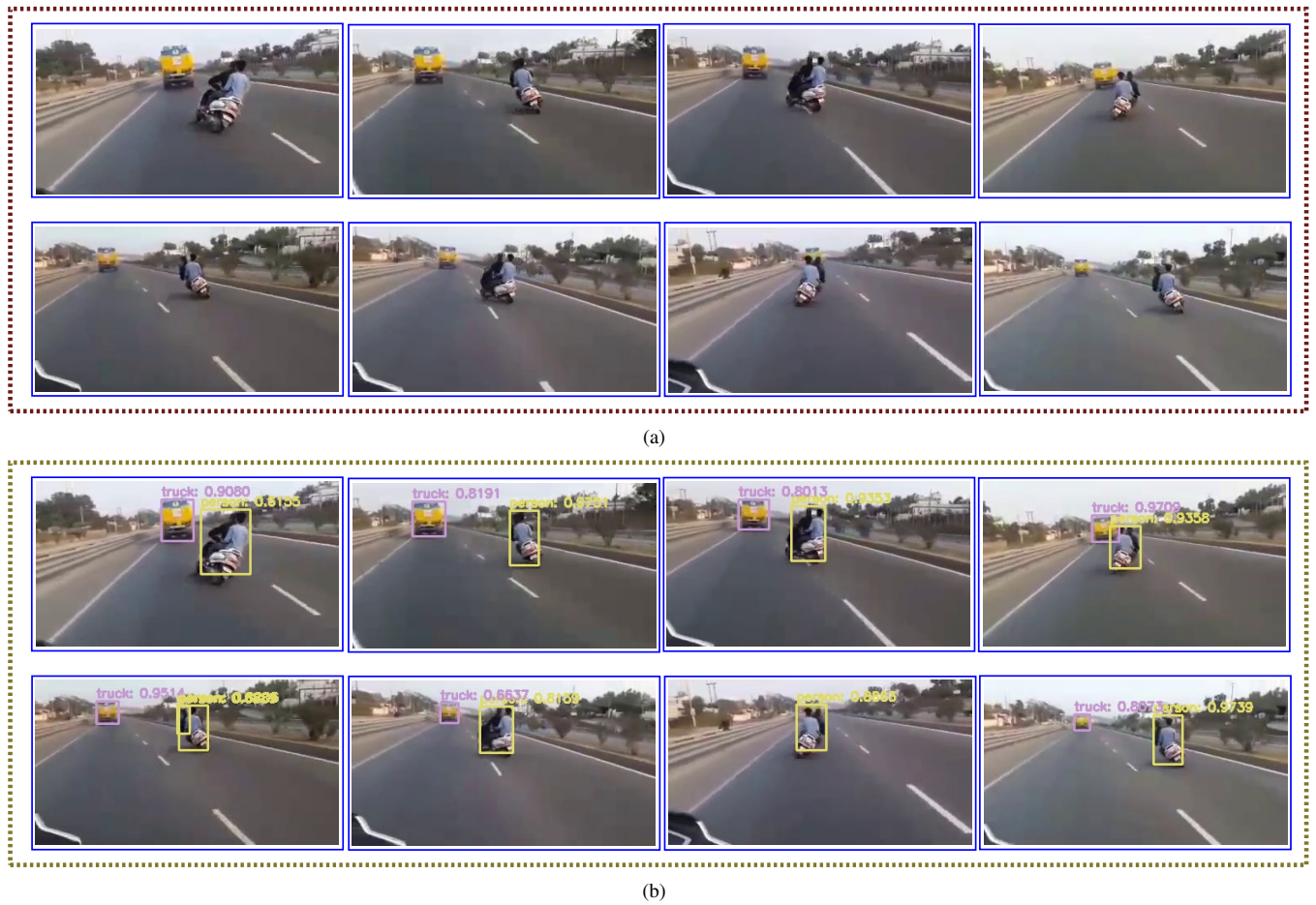


Fig. 6. (a) The extracted frameset of the selected road traffic video. We have taken a few YouTube videos to validate our proposed model. (b) We have used YOLOv3 ([18]) in our proposed model to identify multiple objects such as vehicles, pedestrians, etc in a given traffic video. The YOLOv3 multiple object detection algorithm is applied to these video frames to identify the distinct object with the confidence scores.

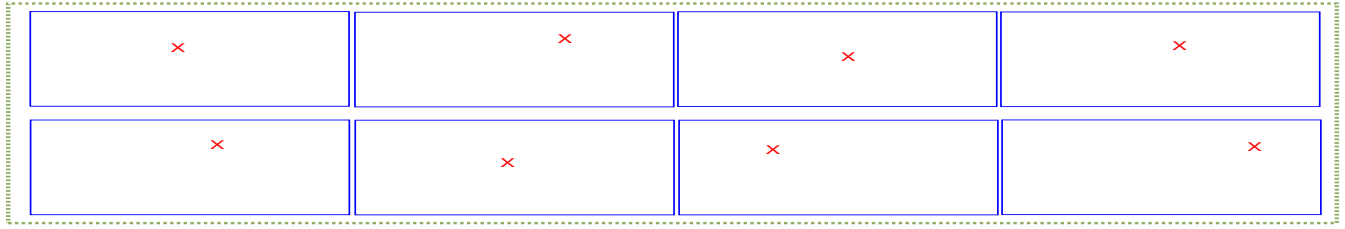


Fig. 7. The detected objects for the given frame, the variable X denotes the object location in the frame. later trajectory is extracted to estimate the θ value.

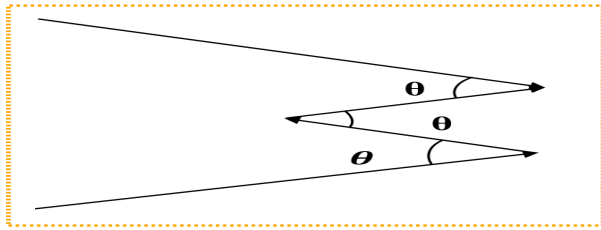


Fig. 8. The trajectory is extracted for the detected objects to estimate the θ value.

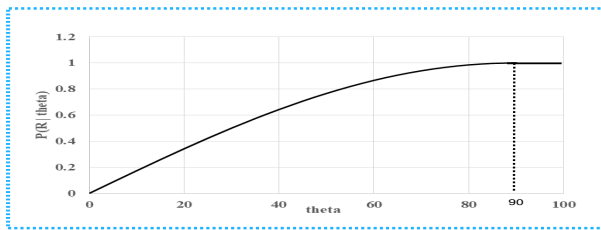


Fig. 9. The function for mapping estimated theta (θ) to probability.

IV. EXPERIMENTS AND RESULTS

A. Dataset

1) *Train dataset* : For the damage detection, we have used a car image dataset from Kaggle ([19]). The dataset has 839 clean and 920 damaged images for training and 230 clean and 230 damaged images for the testing. The available data is quite less and it is known that augmentation of the dataset may help to improve the classifier performance. Therefore we enlarged the available dataset approximately six times by appending it with random rotations (between -20 to 20 degrees) as well as horizontal flip transformations. Table I describes the size of our train and test sets.

2) *Test dataset* : To evaluate the proposed system, purposefully we have taken a **YouTube** short videos of collisions. We have clipped the given videos just before the collision happened and label all the videos as 1. Similarly, we have taken the short videos of the road traffic where no collisions are happened and labeled it as 0. Likewise, we formed the test dataset which contains 72 videos in total and of which 42 are labeled as 1 and the remaining 30 are labeled as 0.

TABLE I

THE DESCRIPTION OF THE IMAGE DATASET WHICH IS USED TO TRAIN AND TEST THE NEURAL NETWORK FOR DENT (DAMAGE ON THE VEHICLE BODY) DETECTION.

	Clean image	Damaged image	Total
Kaggle images for training	839	920	1759
Kaggle images for testing	230	230	460
Augmented images for training	4866	5240	10106
Augmented images for testing	1334	1357	2691
Total images in the dataset			15016

TABLE II

THE CONFUSION MATRIX FOR DAMAGE DETECTION. IN ORDER TO VALIDATE THE CREDENTIAL OF THE DENT DETECTION MODULE, WE HAVE TAKEN KAGGLE DATASET ([19]). IN OUR EXPERIMENT, WE SUCCEEDED TO CLASSIFY THE 218 IMAGES OUT OF 230 DAMAGED CAR IMAGES AND 209 OUT OF 230 CLEAN CAR IMAGES PROPERLY, I.E, THE PROPOSED MODEL GAINS THE ACCURACY OF 92.8% FOR THE DENT DETECTION MODULE.

N=460	Predicted (Clean)	Predicted (Damaged)	
Actual (clean)	TN=209	FP=21	230
Actual(damage)	FN=12	TP=218	230

B. Results

At first, we conducted experiments for module 1 and evaluated the performance. To evaluate the module 1 we have used dataset mentioned in section IV-A1 for which results are shown in Table II. Table III summarizes the obtained result of the dent detection. It can be easily noticed that the data augmentation helps to improve the generalization and achieves better accuracy than just original dataset. We evaluated the proposed system using test data set mentioned in section IV-A2 for which results are shown in Table IV.

As per our observation, we found out that when there is no collision in the video still the model is predicting for the collision. By carefully observing the video we found that even though a collision has not happened the trajectory taken by the vehicle is not smooth which could be the reason for the wrong prediction. It is a narrow escape by the collision by the driver. We observed that the VGG16 model has successfully extracted features that can with very high accuracy classify the dent/damaged vehicle in a given frame. The overall performance is quite promising and acceptable for detecting damaged vehicles using the extracted features by VGG16 model.

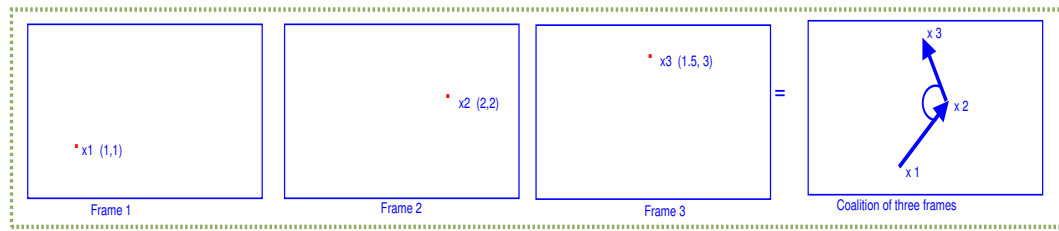


Fig. 10. The estimation of the theta (θ) for the extracted trajectory for the given three points x_1 , x_2 , x_3 . For illustration let $\vec{a} = (1,1)$ and $\vec{b} = (-0.5,1)$, then $\theta = \cos^{-1}(\frac{\vec{a} \cdot \vec{b}}{||\vec{a}|| \cdot ||\vec{b}||}) = \cos^{-1}(0.3162) \approx 72$.

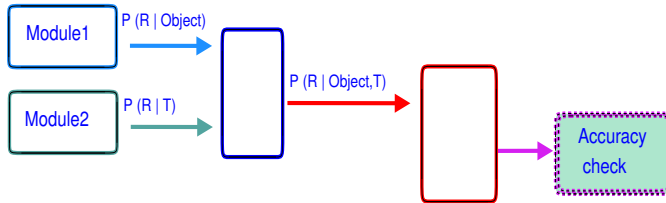


Fig. 11. Verification of the rash driving detection module. For the given traffic, video objects are identified using the YOLOv3 algorithm later probability of rash driving given object $P(R|Object)$ and the probability of rash driving given trajectory $P(R|T)$ is calculated using module 1 and module 2 respectively. By considering the results of these two modules the probability $P(R|Object, T)$ of rash driving is predicted.

TABLE III
THE TEST ACCURACY FOR TRAINING THE MODEL (DENT DETECTION).

Training data set	Accuracy
Original image (Without Augmentation)	92.81
Augmented image	96.17

V. CONCLUSION

In this paper, we have described a visual analysis method to characterize driving behavior, with a specific focus on rash driving. Our system is based on calibrated single-camera videos. The main aim of this system is to detect rash driving on highways using computer vision which is cheaper and easy to install. A number of collisions on highways increase day by day so it is necessary to monitor the rash driving behavior of the driver passing on highways so as to reduce the collision cases. It also reduces the difficulties of the traffic police department and makes it easy to control the rash driving on highways without human intervention by detecting them and can take legal action to avoid such incidents. This concept can be enhanced in the future by integrating multiple cameras that could capture the video in multiple views, to enhance the accuracy and to give more precise detection. In the future, it

TABLE IV
THE PROPOSED SYSTEM PERFORMANCE STATISTICS.

	collision	No collision
collision	80%	20%
No collision	60%	40%

can also store the data (number of times a vehicle detected in rash driving events) and can determine the extent of threat for that vehicle.

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